

Ali Adil Waseem

AI Engineer | Python Developer | Automation & Backend Systems

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PROFESSIONAL SUMMARY

AI Engineer specializing in the design and deployment of autonomous agent systems, multi-step AI pipelines, and workflow orchestration. Experienced in building production-ready agentic applications powered by local and cloud LLMs, with a focus on systems that act, self-correct, and scale, not just respond. Proven ability to architect end-to-end intelligent workflows from voice interfaces to browser automation, document intelligence, and backend orchestration. Comfortable working at the intersection of LLM reasoning, tool use, and real-world system integration. Experienced in building GraphRAG-powered memory systems and RAG pipelines using vector databases, combining retrieval-augmented generation with agentic architectures for context-aware, locally-run AI products.

EDUCATION

2025 **Bachelor of Science in Artificial Intelligence**
Lahore, Pakistan University of Management and Technology (UMT), Lahore, Pakistan

TECHNICAL SKILLS

AI / ML

- PyTorch, TensorFlow, NLP, OCR
- Computer Vision, Speech Recognition
- Text Summarization, Object Segmentation
- Model Inference Optimization

Programming

- Python, JavaScript, HTML, CSS
- Next.js, React

Specialized Models

- SAM 2.1, Real-ESRGAN, BigLaMa
- MediaPipe
- RoBERTa, BART, sentence-transformers
- LoRA Fine-tuning, Model Chaining, Multi-model Pipelines

Automation & Web

- Playwright, Selenium, BeautifulSoup
- Web Scraping, Browser Automation
- Desktop Automation, System Integration

Agent & Orchestration

- Agentic Pipelines, Multi-step Workflow Design, Tool Use & Function Calling
- Local LLM Integration (Ollama, Llama 3.1)
- LLM APIs | Claude API, Gemini API, Groq
- Prompt Engineering

Frameworks & Tools

- FastAPI, Streamlit, PySide6, REST APIs
- Docker, Git, OpenCV, MediaPipe

RAG & Vector Systems

- GraphRAG, RAG Pipelines, Semantic Search
- Vector Embeddings, LanceDB, all-MiniLM
- Knowledge Graphs, Reranking

PROJECTS

NEXA - AI Desktop Assistant

Personal Project

- Architected a voice-driven AI desktop assistant with 190+ callable functions, enabling natural language orchestration of Windows workflows, productivity tasks, and system control with approximately 95% intent accuracy.

- Built a local GraphRAG memory system where personal memories are stored as a relational node graph with explicit relationships, semantically retrieved via all-MiniLM embeddings and LanceDB to ground every response in personal context.
- Designed an intent-routing pipeline that maps natural language commands to the correct automation function, functioning as a production-grade agent dispatcher with online/offline LLM switching via Ollama.
- Visualized the memory graph through an interactive node-based UI panel, giving users full visibility and control over NEXA's knowledge state and relationship map.
- Built a content refinement agent with 16 AI-powered transformation tools, demonstrating multi-step reasoning, output chaining, and modular function design.
- Engineered end-to-end agent loops covering input, intent classification, tool dispatch, execution, and response, mirroring production agentic workflow architecture.

Cortx - Autonomous Web Research Agent

Personal Project

- Built a fully local autonomous web research assistant with a polished PyQt6 desktop GUI, decomposing user queries into intent, discovering sources, scraping the web, and delivering source-cited answers end-to-end without human intervention.
- Implemented a multi-tier scraping engine with automatic fallback across httpx, Playwright, and Selenium, handling JavaScript-rendered pages and anti-bot protected sites reliably at scale.
- Architected a RAG pipeline using LanceDB vector indexing to embed and cache scraped content, enabling intelligent follow-up queries without re-scraping and reducing response latency significantly.
- Orchestrated a LangChain LCEL pipeline with modular chains for intent extraction, query generation, and answer synthesis, maintaining clean separation of concerns across the research workflow.
- Integrated Ollama with Llama 3.1 as the primary offline reasoning core and Groq cloud as a graceful fallback, ensuring full functionality with zero API cost and complete privacy.
- Designed explicit context gating for follow-up queries, where context is only used when intentionally selected, giving users precise control over research continuity and answer scope.

TrueFit - AI Body Measurement Extraction System

Final Year Project | University of Management and Technology, Lahore

- Tech Stack: Python, Computer Vision, MediaPipe, FastAPI, Docker | github.com/OK-ALI/TrueFit
- Developed an AI pipeline extracting 15+ anthropometric measurements from a single image using MediaPipe pose estimation with 33-landmark detection.
- Designed multi-method measurement algorithms using anthropometric modeling, symmetry validation, and cross-validation for reliability.
- Built production-ready REST API and CLI interfaces using FastAPI with Docker-based containerized deployment.
- Automated structured output generation: JSON reports, HTML summaries, and annotated image exports.

NoteWise - AI Document Intelligence Assistant

Personal Project

- Tech Stack: Python, Streamlit, Gemini 2.0 Flash, OCR, NLP | github.com/OK-ALI/NoteWise
- Built a document intelligence pipeline that ingests handwritten notes, PDFs, and presentations, then generates structured summaries and study materials via AI.
- Developed multi-mode AI summarization with key-insight extraction and automatic question generation (MCQs, True/False, short-answer).
- Designed Streamlit-based interface for zero-friction educational use.

NEXUS - AI Image Editing Pipeline

Personal Project

- github.com/OK-ALI/Nexus-2.0
- Tech Stack: Python, SAM 2.1, Big-LaMa, Real-ESRGAN, Computer Vision
- Built a multi-model AI pipeline (SAM 2.1 -> Big-LaMa -> Real-ESRGAN) for object removal and image restoration, demonstrating chained model orchestration.
- Implemented bounding-box-prompted segmentation for precise object selection followed by inpainting and optional super-resolution enhancement.

LEADERSHIP & INVOLVEMENT

03/2025 – 08/2025
Lahore, Pakistan

- President**
- Led club operations, events, and a team of student coordinators — building cross-functional collaboration and execution habits.
 - Mentored new members and drove participation through structured community-building initiatives.